1. You are working with a newly discovered mutagen, and you wish to determine the base change that it introduces into DNA. Thus far, you have determined that the mutagen chemically alters a single base in such a way that its base-pairing properties are altered permanently. To determine the specificity of the alteration, you examine the amino acid changes that take place after mutagenesis. A sample of what you find is shown here:

| | |
|---|---|
| Original: | Gln–His–Ile–Glu–Lys |
| Mutant: | Gln–His–Met–Glu–Lys |
| Original: | Ala–Val–Asn–Arg |
| Mutant: | Ala–Val–Ser–Arg |
| Original: | Arg–Ser–Leu |
| Mutant: | Arg–Ser–Leu–Trp–Lys–Thr–Phe |

Ile: AUU, AUC, AUA — Met: AUG

Asn: GAU, GAC — Ser: UCU, UCC, UCA, UCG

stop: UAA, UAG, UGA — Trp: UGG

∴ mRNA : A → G

DNA : T → C

2. Some mutagens cause genetic changes that can be "corrected" by reexposing cells to the same mutagen. Other mutagens do not behave in this way. Provide one example of each of these two types of agents and describe the mutational changes caused in DNA. Explain why some mutagens behave in one way, while others do not.

① 亚硝酸导致的碱基替换可以被导致相同种类的突变的mutagen修复。

② X射线导致的染色体结构突变不能被mutagen修复，因为通常是碱基缺失或translocation等的突变。